

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |

Kent, OH - F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) specializing in healthcare AI agent implementations, RAG systems, and end-to-end deployment of LLM-powered workflows in HIPAA-conscious environments. Proven ability to manage AI agent deployments for complex healthcare use cases - improving contextual retrieval precision by ~35%, reducing hallucinations by >30%, and delivering ~30% fewer post-deployment defects. Experienced in building integrations, structuring clinical knowledge for automated systems, and optimizing agentic workflows from data ingestion through production API serving. Ready to serve as the technical bridge between Assort Health's AI platform and healthcare client environments.

CORE COMPETENCIES

Healthcare AI Agent Deployment • RAG Systems & Retrieval Optimization • Agentic LLM Workflows • Python & FastAPI • Docker & Production Packaging • API Integration & System Connectivity • HIPAA-Conscious Data Handling • Prompt Engineering & Guardrails • LLMOps & Evaluation Pipelines • Customer-Facing Technical Delivery • LangChain & LlamaIndex • Clinical Knowledge Structuring

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology & AI Consulting

- Managed end-to-end deployment of **RAG systems for clinical knowledge retrieval** - recursive semantic chunking, transformer-based embeddings, retrieval evaluation; improved contextual retrieval precision by ~**35%** and reduced irrelevant retrieval by >**30%**; retrieval-grounded response alignment exceeded **90%**.
- Built and optimized **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~**25%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall by **15-20%** on high-risk categories while holding precision >**90%**.
- Packaged ML/LLM inference as **FastAPI/Flask RESTful services** containerized with Docker; structured logging and load simulation reduced post-deployment defects by ~**30%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and system-limitation docs for stakeholders.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech & Applied NLP

- Built a **transformer-based video summarization system** across 5,000+ recorded sessions; reduced manual review time by **60-70%** (hours → <5 minutes per video); achieved ~**85% highlight alignment** with human curation.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~**30%** and raised contextual answer accuracy >**85%**.
- Deployed full stack as a **Flask-based API** with lightweight web interface for non-technical users.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- Built **CI/CD pipelines with Jenkins** for SQL schema deployments; reduced deployment errors >**30%** and improved release cycle time ~**35-40%**.
- Designed performance dashboards (SQL Server DMVs + Grafana) to track query health; cut incident recurrence ~**25%**.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, FastAPI, vector search, multi-agent orchestration, healthcare data standards

Manga Lens - Chrome Extension for Real-Time AI Translation Shipped

Full-stack browser extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider payload handling, seven-day translation cache, per-domain selector configs for 29 sites. Demonstrates rapid integration across heterogeneous AI APIs under production constraints.

TypeScript, Manifest V3, multi-provider API integration, caching, browser automation

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with **unified YOLOv8**; reduced inference latency ~**30%**. Sliding-window confidence aggregation suppressed blink-driven false positives ~**25%**; adaptive frame skipping for stable real-time operation.

YOLOv8, OpenCV, Python, real-time inference optimization

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL (T-SQL, PostgreSQL, SQLite)

AI Agent & LLM Stack: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, LangChain, LlamaIndex

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers

Infrastructure & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Workflow Automation, Computer Vision, Multimodal Systems